

Welcome to **Troubleshooting O2 Jobs!**
We will get started in a few minutes.

Slides available at
github.com/hmsrc/user-training
TroubleshootingO2Jobs.pdf



Troubleshooting O2 Jobs

kathleen_chappell@hms.harvard.edu

Some slides are courtesy of
william_rodriguez@hms.harvard.edu

HMS Research Computing:
rhelp@hms.harvard.edu

Information Technology - Research Computing
Harvard Medical School
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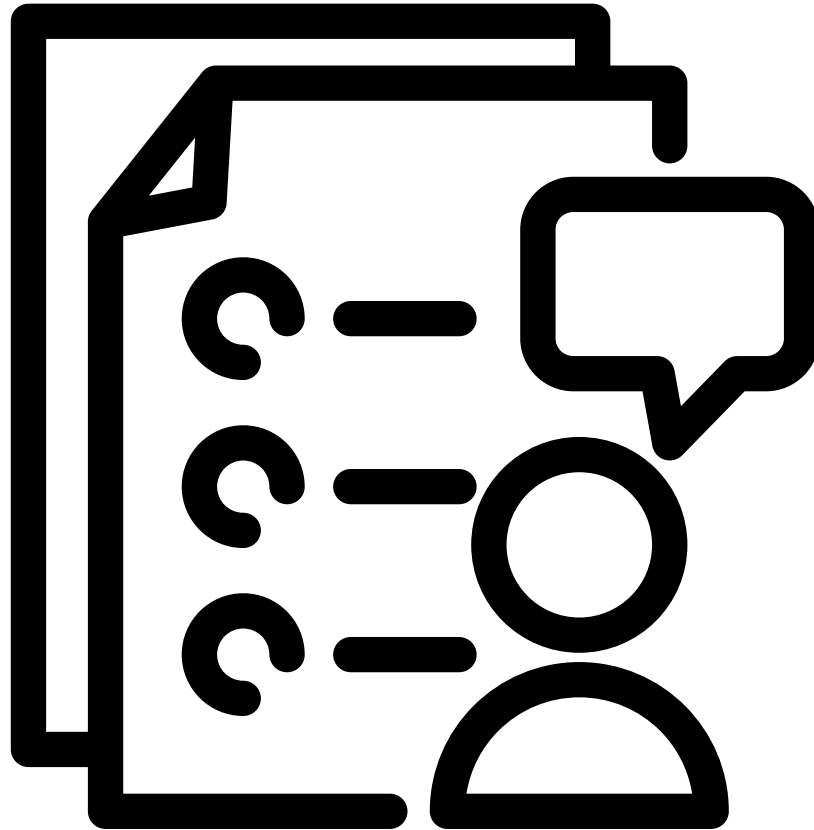
Class Logistics

- Put questions in chat, or unmute and ask
- Slides available at github.com/hmsrc/user-training
 - TroubleshootingO2Jobs.pdf
- We will have hands-on exercises. If you do not have an O2 account, we will have training accounts (that only work for the class duration) available
- Comments/feedback welcome at course survey
- Resource for after the class: [Troubleshooting Jobs wiki page](#)

Who is this class for?

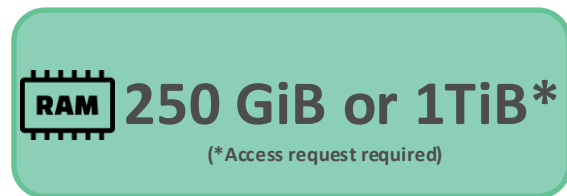
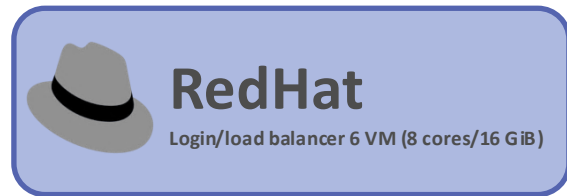
- Folks who have taken “Intro to O2” or have already submitted jobs on the O2 cluster
 - and who want to ***become more independent in debugging their work.***
- We assume a baseline amount of knowledge about how the cluster works, but we will have a quick review.
- We will focus on troubleshooting jobs on the command line.

Poll

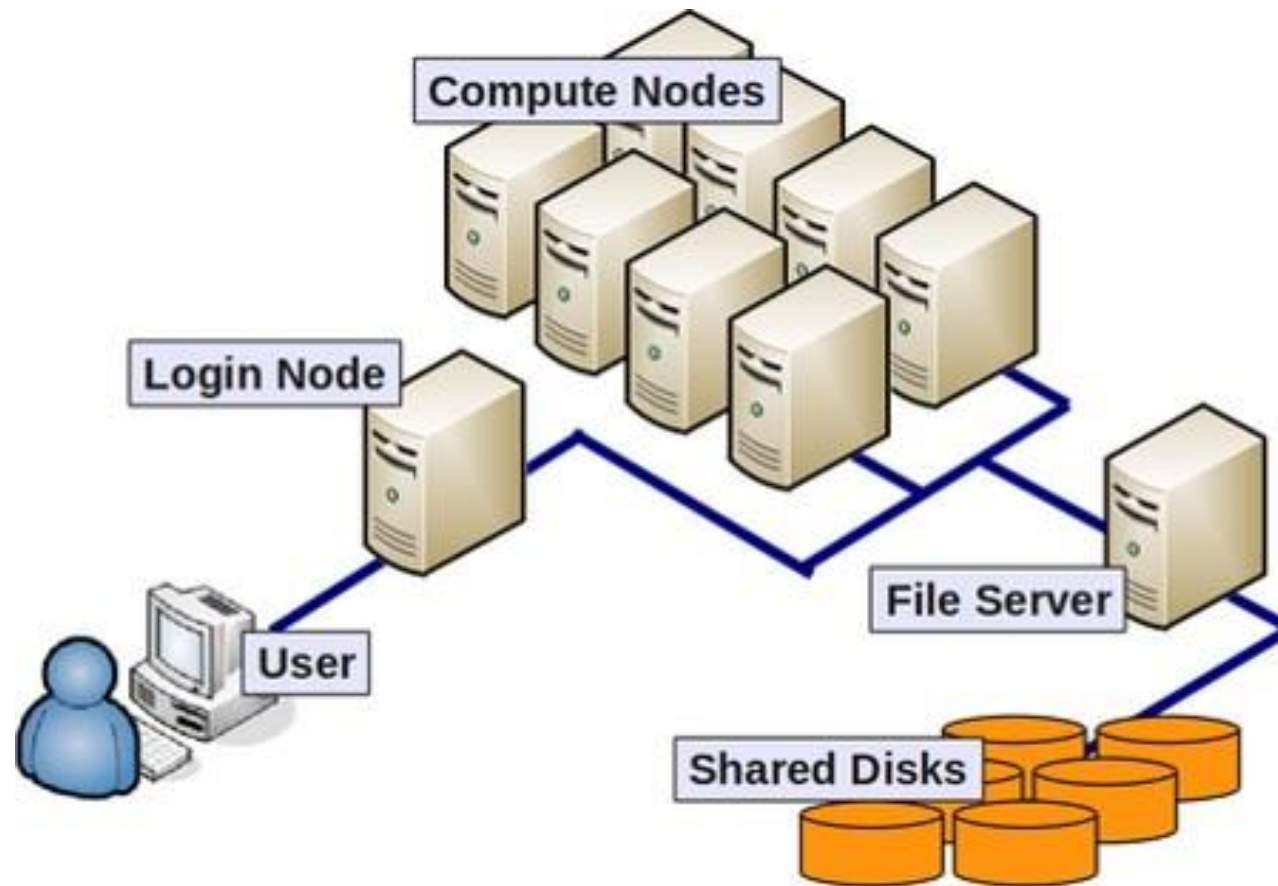


What is O2?

- One of HMS Research Computing's High-Performance Compute clusters
- SLURM scheduler

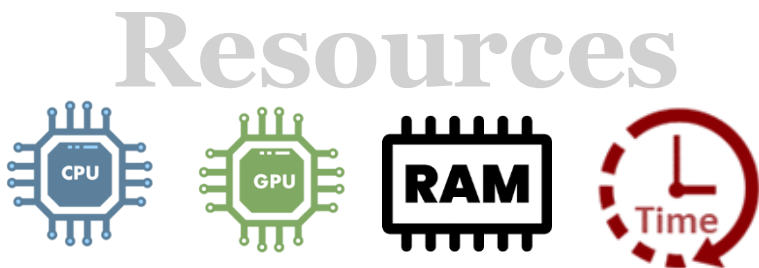


Generic Cluster Architecture



Job submissions

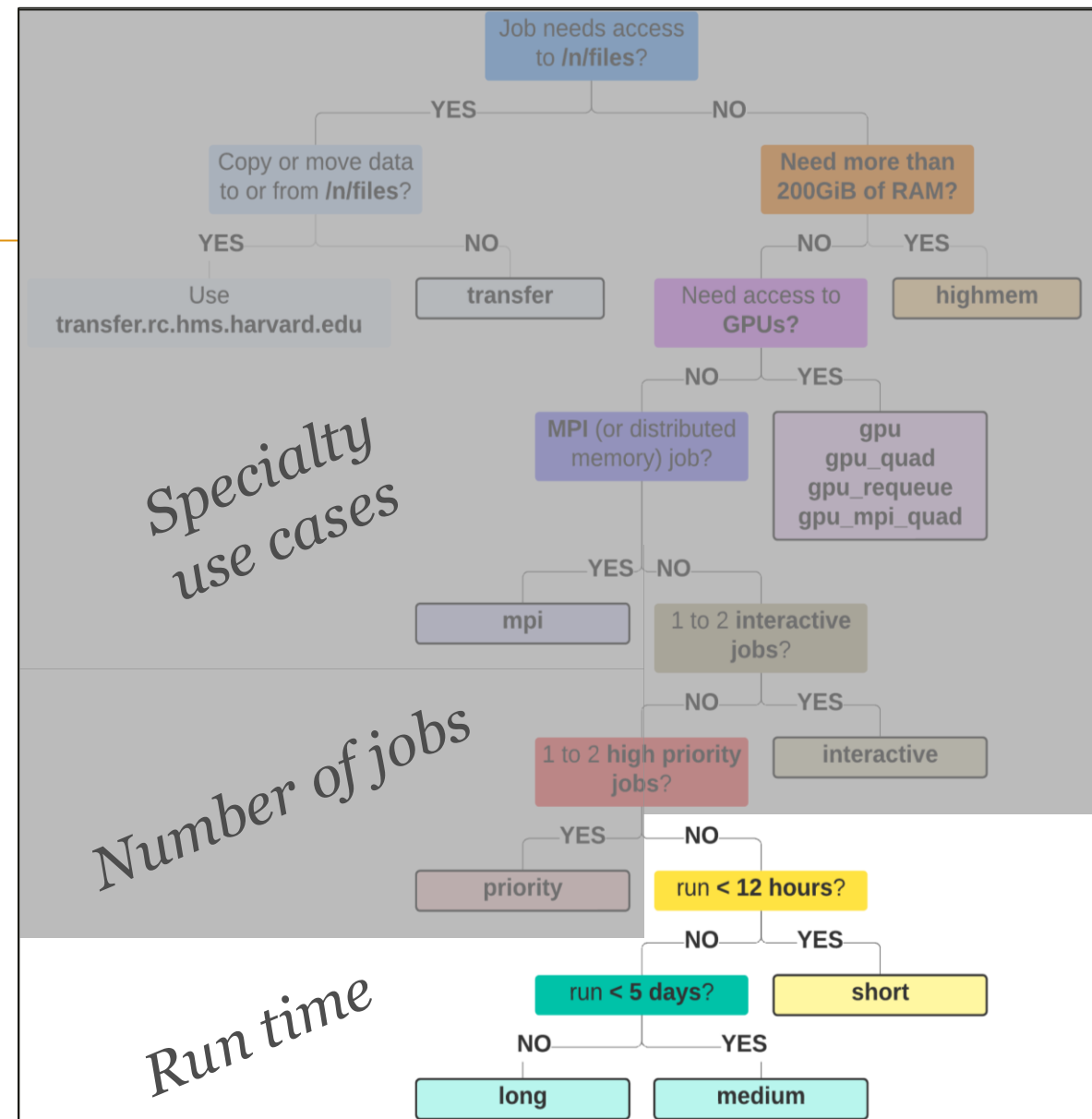
*A job is a **resource allocation** for a user for a **set amount of time**.*



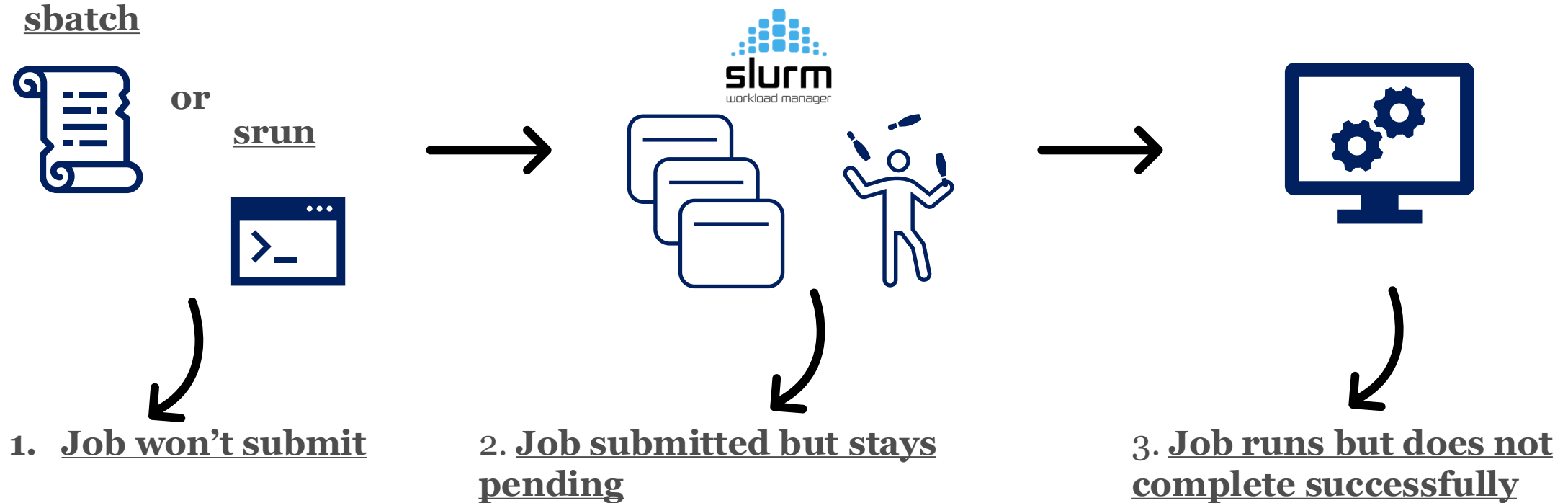
	<u>srun</u> interactive	<u>sbatch</u> batch
Requires a script?	No	Yes
Run analyses in real-time on the compute node(s)?	Yes	No
Job ends if you disconnect?	Yes	No

Choosing a Partition

- Narrow down a partition to use:
 - If you have a specialty use case (e.g. /n/files access, high memory, GPU, MPI, etc.)
 - Number of jobs to run
 - How long your job needs to run



Three main types of problems with jobs:



How to debug: (1) *Jobs that won't submit*

- Read error messages!

- Missing command

```
$ srun --pty -p interactive -t 0-2  
srun: fatal: No command given to execute.
```

- Missing partition

```
$ srun --pty -t 0-2 bash  
srun: error: Job not submitted: please specify partition with -p.  
srun: error: Unable to allocate resources: Invalid partition name specified
```

- Missing walltime

```
$ srun --pty -p interactive bash  
srun: error: Job not submitted: please specify runtime limit with -t.  
srun: error: Unable to allocate resources: Requested time limit is invalid (missing or exceeds some limit)
```

- Missing --account

```
$ srun --pty -p interactive -t 0-2 bash  
srun: error: lua: You have more than one Slurm account, must specify an account to submit  
srun: error: cli_filter plugin terminated with error
```



How to debug: (2) *Jobs that won't start*

- Check [O2squeue](#) (pending/running jobs)
 - For pending jobs, the “NODELIST(REASON)” will say why your job is not running

```
[kmk34@compute-a-16-160 ~ 2024-04-19 15:43:42]$ O2squeue
```

JOBID	PARTITION	STATE	TIME_LIMIT	TIME	NODELIST(REASON)	ELIGIBLE_TIME	START_TIME	TRES_ALLOC
36645028	interactive	PENDING	4:00:00	0:00	(Resources)	2024-04-19T15:37:09	2024-04-19T17:49:38	cpu=1,mem=250G,node=1,billing=16
36645882	interactive	PENDING	10:00	0:00	(Priority)	2024-04-19T15:43:33	N/A	cpu=1,mem=4G,node=1,billing=1
36644759	interactive	RUNNING	2:00:00	14:06	compute-a-16-160	2024-04-19T15:28:10	2024-04-19T15:29:37	cpu=1,mem=4G,node=1,billing=1
36645850	short	PENDING	12:00:00	0:00	(None)	2024-04-19T15:43:15	N/A	cpu=1,mem=250G,node=1,billing=16

```
[kmk34@compute-a-16-160 ~ 2024-04-19 15:48:00]$ O2squeue
```

JOBID	PARTITION	STATE	TIME_LIMIT	TIME	NODELIST(REASON)	ELIGIBLE_TIME	START_TIME	TRES_ALLOC
36645028	interactive	PENDING	4:00:00	0:00	(QOSMaxJobsPerUserLimit)	2024-04-19T15:37:09	N/A	cpu=1,mem=250G,node=1,billing=16
36644759	interactive	RUNNING	2:00:00	18:24	compute-a-16-160	2024-04-19T15:28:10	2024-04-19T15:29:37	cpu=1,mem=4G,node=1,billing=1
36645882	interactive	RUNNING	10:00	4:09	compute-a-16-166	2024-04-19T15:43:33	2024-04-19T15:43:52	cpu=1,mem=4G,node=1,billing=1
36646078	long	PENDING	30-00:00:00	0:00	(None)	2024-04-19T15:47:22	N/A	cpu=20,mem=250G,node=1,billing=35

- [Reference here](#) for a list of “REASONS” why a job is pending



How to debug: (3) *Jobs that run but fail*

- Check [O2 jobs report](#) (jobs that have finished running)
 - Look at “STATE” column & “EFF” columns for “CPU”, “RAM” and “WALLTIME”

```
[[kmk34@login02 ~ 2024-04-19 15:20:23]$ O2_jobs_report -s 2024-04-01
```

JOBID	USER	ACCOUNT	PARTITION	STATE	STARTTIME	WALLTIME(hr)	RUNTIME(hr)	nCPU, RAM(GB), nGPU	PENDINGTIME(hr)	CPU_EFF(%)	RAM_EFF(%)	WALLTIME_EFF(%)
36454631	kmk34	rccg	interactive	TIMEOUT	2024-04-16	2.0	2.0	1, 10.0, 0	0.03	3.8	32.6	100.2
36511328	kmk34	rccg	short	TIMEOUT	2024-04-17	1.0	1.0	1, 1.0, 0	0.04	0.1	0.7	100.2

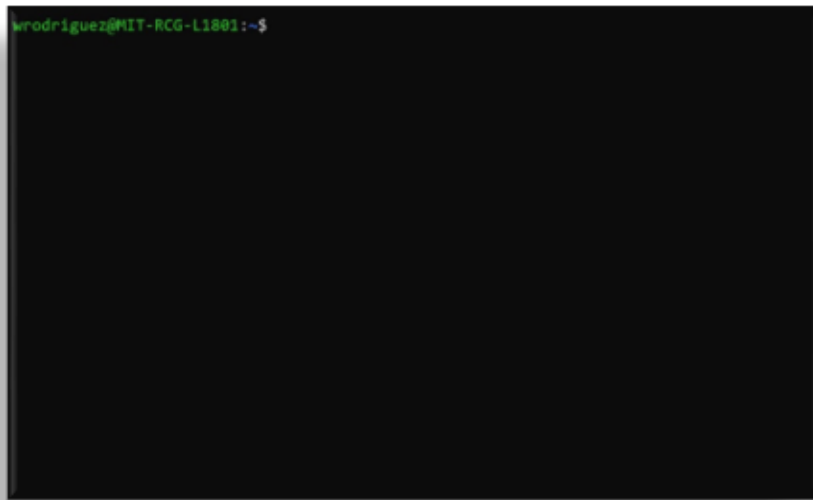
- [Reference here for a list of “STATES” that a job could end in](#)
- Check error and output files
 - Interactive jobs have messages printed to screen
 - Batch jobs have error/output redirected to file(s)
 - See filenames specified in –e and –o in your job script OR will be written to slurm-jobid.out



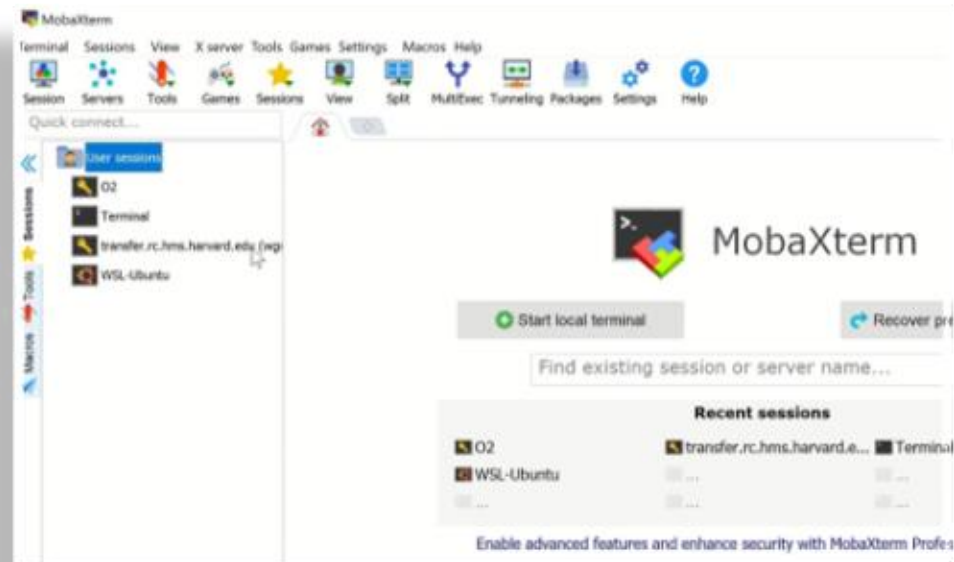
Login to the O2 cluster

ssh hmsid@o2.hms.harvard.edu
replace hmsid with your actual hmsid

Mac/Linux: Terminal



Windows: MobaXterm



If you do not have an O2 account, use a training account!



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Exercises

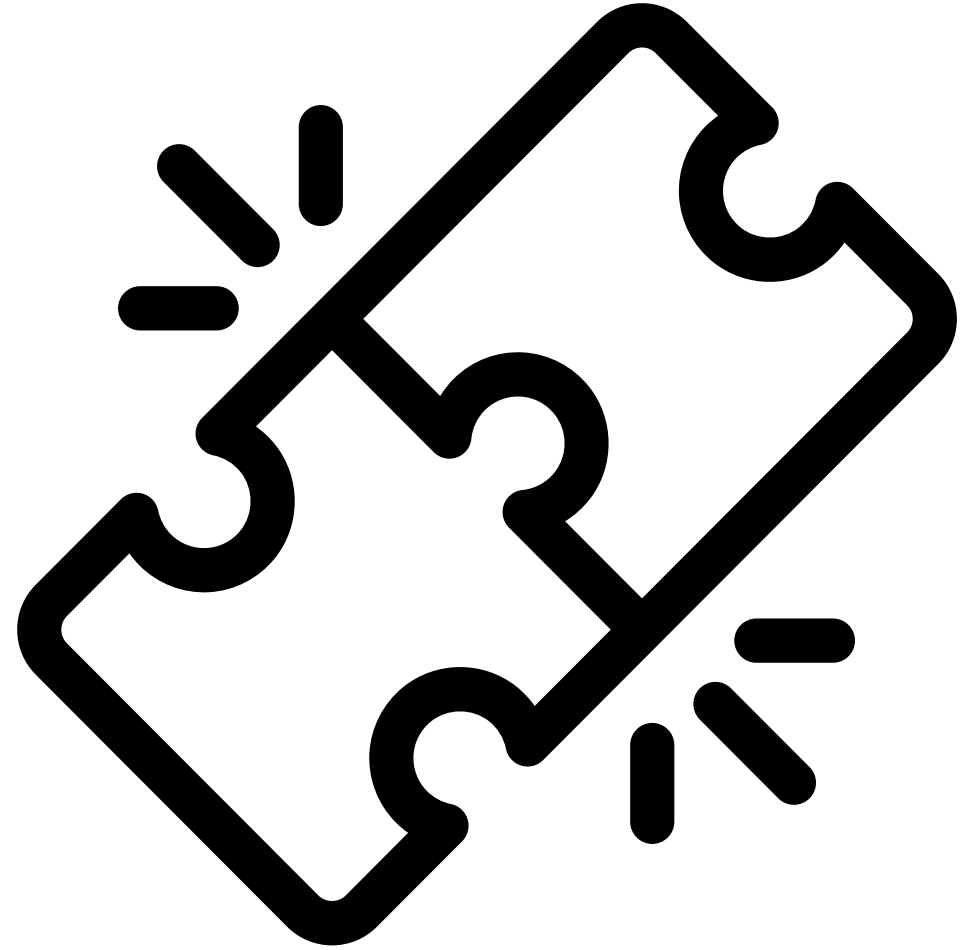


Image credit: [Problem Solving](#) by [Symbolon](#) from [Noun Project](#) (CC BY 3.0)



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Information Technology 15

Copy class materials

```
$ cp -r /n/groups/rc-training/troubleshooting_jobs/ ~
```

```
$ cd troubleshooting_jobs/
```

```
├── example_proteins.fasta
├── exercise1.sh
├── exercise2.sh
├── exercise3.sh
└── solutions
    ├── exercise1.sh
    ├── exercise2.sh
    └── exercise3.sh
```

The exercise scripts will fail on purpose, so we can debug them.

There are “corrected” scripts in the solutions/ subdirectory for your future reference.



Exercise 1: job won't submit!

```
$ sbatch exercise1.sh
```

```
#!/bin/bash
#SBATCH -p short           # Partition to submit to
#SBATCH -t 1-0             # Time in days-hours format
#SBATCH -c 1               # Number of cores requested
#SBATCH --mem=1G           # Memory total in GB (see also --mem-per-cpu)
#SBATCH -o hostname.%j.out # Standard out goes to this file
#SBATCH -e hostname.%j.err # Standard err goes to this file

hostname
```

- Read the error message when submitting the job.

```
[kmk34@login06 troubleshooting_jobs 2024-04-23 13:49:22]$ sbatch exercise1.sh
sbatch: error: Batch job submission failed: Requested time limit is invalid (missing or exceeds some limit)
```

Exercise 2: job won't dispatch

```
$ sbatch exercise2.sh
```

```
#!/bin/bash
#SBATCH -p gpu                # Partition to submit to
#SBATCH -t 0-0:05             # Time in days-hours:minutes format
#SBATCH -c 1                  # Number of cores requested
#SBATCH --mem=1G               # Memory total in GB (see also --mem-per-cpu)
#SBATCH -o gpu_job.%j.out     # Standard out/err goes to this file

nvidia-smi
```

- Check O2squeue for the REASON why the job won't dispatch

```
[kmk34@login06 troubleshooting_jobs 2024-04-22 17:06:49]$ O2squeue
```

JOBID	PARTITION	STATE	TIME_LIMIT	TIME	NODELIST(REASON) (QOSMinGRES)	ELIGIBLE_TIME	START_TIME	TRES_ALLOC
36777815	gpu	PENDING	5:00	0:00		2024-04-22T17:02:34	N/A	cpu=1,mem=1G,node=1,billing=1

Exercise 3: job runs, but fails

```
$ sbatch exercise3.sh
```

```
#!/bin/bash
#SBATCH -p short
#SBATCH -t 0-0:15
#SBATCH -c 1
#SBATCH --mem=1G
#SBATCH -o submits_&_fails.%j.out

colabfold_search \
--db-load-mode 2 \
--mmseqs mmseqs \
--use-env 1 \
--use-templates 0 \
--threads 4 \
/home/$USER/troubleshooting_jobs/example_proteins.fasta /n/shared_db/misc/mmseqs2/14-7e284
/home/$USER/troubleshooting_jobs/
```

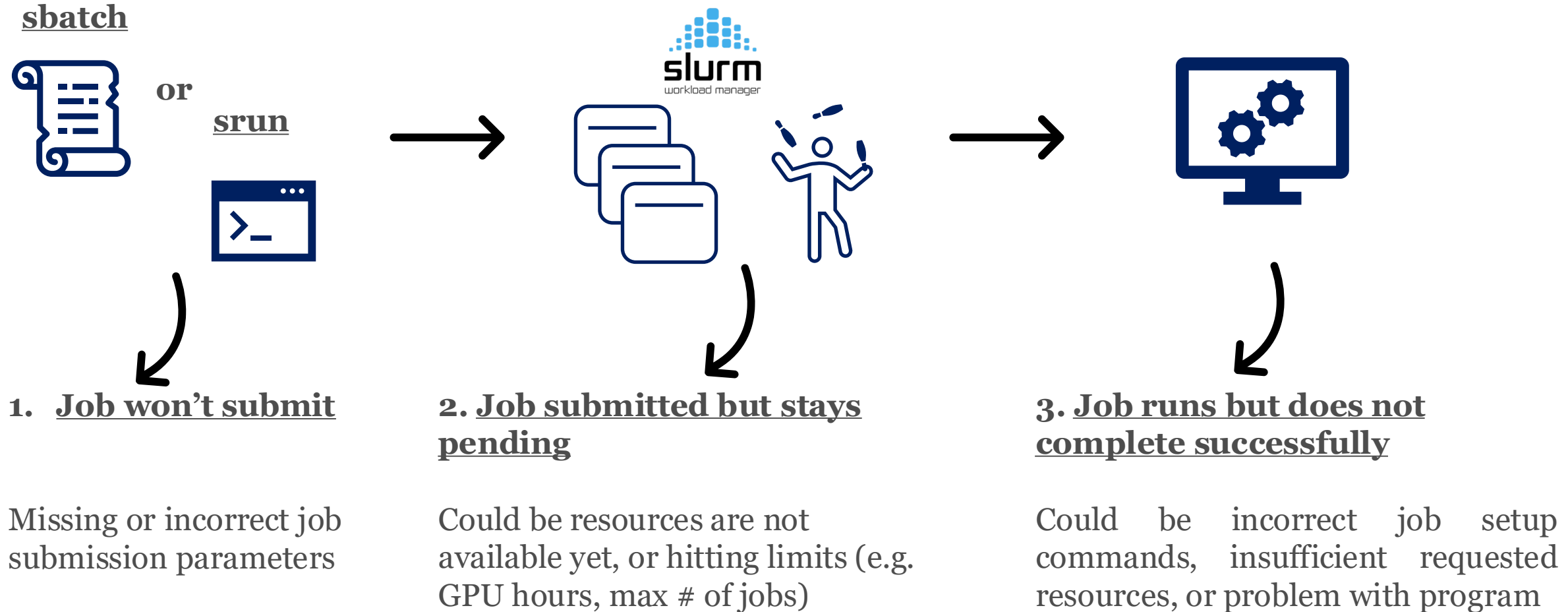
Partition to submit to
Time in days-hours:minutes format
Number of cores requested
Memory total in GB (see also --mem-per-cpu)
Standard out/err goes to this file

- Check O2_jobs_report and out/err file

Revisiting a previous slide:



Three main types of problems with jobs:



An additional type of job problem



Check your job efficiency by

Reading weekly job usage report sent to your email!

Or use [O2 jobs report --report](#)

4. Job runs but is inefficient

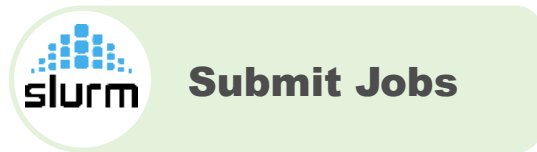
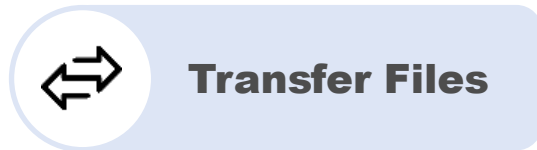
Requests more resources than are required, for example:

- Multiple cores for single threaded workflow
- 6 hours when the job finishes in 30 minutes
- 50GiB of RAM when the maximum used is 10GiB

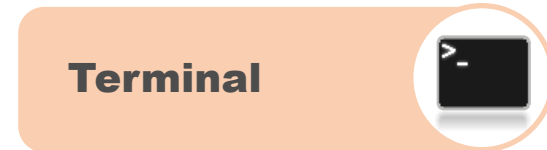
Fix by reducing job resource requests



O2 Portal: another way to access O2



 **Portal**



o2portal.rc.hms.harvard.edu

[O2 Portal docs](#) | [O2 Portal class slides](#)



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Debugging O2 Portal Jobs

 My Interactive Sessions

“My Interactive Sessions” is to the right of HMS-RC Applications in the top navigation bar

RStudio Environment (36511328) 1 node | 1 core | Running

Host: compute-a-16-162.o2.rc.hms.harvard.edu Delete

Created at: 2024-04-17 14:02:44 EDT

Time Remaining: 59 minutes

Session ID: 208113a2-97c7-4911-b9ab-1d93620ce236

Progress & Error log: [output.log](#)

Ⓜ Connect to RStudio Server

Click on the `output.log` link in the session card.

Check for any error messages.

Server to open port 16890...

ules:

8) glib/2.72.1	15) xz/5.2.5
9) libxml/2.9.11	16) boost/1.75.0
10) freetype/2.11.1	17) gsl/2.7.1
11) libffi/3.4.2	18) hdf5/1.12.1
12) fontconfig/2.13.94	19) glpk/5.0
13) harfbuzz/3.3.2	20) R/4.2.1
14) cairo/1.16.0	21) git/2.35.3

```
compute-a-16-162.o2.rc.hms.harvard.edu
/etc/krb5.conf,/tmp/tmp.98LqbRo80r:/tmp,/n,/home/kmk34/ondemand/data/sys/dashboard/
batch_connect/sys/RC_rstudio/output/208113a2-97c7-4911-b9ab-1d93620ce236/etc/rsession.conf:/
etc/rstudio/rsession.conf,/home/kmk34/ondemand/data/sys/dashboard/batch_connect/sys/RC_rstudio/
output/208113a2-97c7-4911-b9ab-1d93620ce236/etc/rsession.conf:/etc/rstudio/rsession.conf,/home/
kmk34/ondemand/data/sys/dashboard/batch_connect/sys/RC_rstudio/output/208113a2-97c7-4911-
b9ab-1d93620ce236/etc/database.conf:/etc/rstudio/database.conf,/home/kmk34/ondemand/data/sys/
dashboard/batch_connect/sys/RC_rstudio/output/208113a2-97c7-4911-b9ab-1d93620ce236/etc/
logging.conf:/etc/rstudio/logging.conf
APPTAINERENV_R_LIBS_USER
R_LIBS_USER
BINDINGS /etc/krb5.conf,/tmp/tmp.98LqbRo80r:/tmp,/n,/home/kmk34/ondemand/data/sys/dashboard/
batch_connect/sys/RC_rstudio/output/208113a2-97c7-4911-b9ab-1d93620ce236/etc/rsession.conf:/
etc/rstudio/rsession.conf,/home/kmk34/ondemand/data/sys/dashboard/batch_connect/sys/RC_rstudio/
output/208113a2-97c7-4911-b9ab-1d93620ce236/etc/rsession.conf:/etc/rstudio/rsession.conf,/home/
kmk34/ondemand/data/sys/dashboard/batch_connect/sys/RC_rstudio/output/208113a2-97c7-4911-
b9ab-1d93620ce236/etc/database.conf:/etc/rstudio/database.conf,/home/kmk34/ondemand/data/sys/
dashboard/batch_connect/sys/RC_rstudio/output/208113a2-97c7-4911-b9ab-1d93620ce236/etc/
logging.conf:/etc/rstudio/logging.conf
```



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Contact us!



Email: rchelp@hms.harvard.edu



Website: <https://it.hms.harvard.edu/rc>



Office hours:

Wednesdays, 1:00-3:00 pm

Zoom: <https://rc.hms.harvard.edu/office-hours>



We welcome your feedback in the survey!

<https://bit.ly/hmsrc-troubleshoot> or



Questions?

